CSA17 – Artificial Intelligence: Assessment Tool 3 (MCQ Solutions)

COURSE: Artificial Intelligence (CSA17) — Assessment Tool 3 CO3: Knowledge Representation and Reasoning (Propositional Logic, FOL, Inference, Resolution, Chaining, Learning) Total Marks: 30 (30 questions × 1 mark each)

> **Mark split:** The paper states 30 questions for 30 total marks, so each MCQ carries **1 mark** — awarded for circling the correct option below. No partial marking applies per the rubric (correctness is binary per question; the rubric's weightage columns — Conceptual Understanding, Accuracy, Application, Time Management, Interpretation — are qualitative bands applied to overall quiz performance, not per-question sub-marks).

Answers

Q	Correct Option	Justification
1	**b)** An agent that uses a knowledge base and logical inference to decide actions	Defines a Logical Agent: it reasons over a KB via inference rather than reacting randomly or following fixed scripts.
2	**d)** Both b and c	¬P ∨ Q is the material-implication equivalence; ¬Q -> ¬P is the contrapositive — both are logically equivalent to P -> Q.
3	**b)** A conjunction of disjunctions (clauses)	CNF = AND of ORs — each clause is a disjunction of literals, and the whole formula is a conjunction of these clauses.
4	**b)** Every entity that is a student passes	∀x: Student(x) -> Passes(x) is a universally quantified conditional: for all x, being a student implies passing.
5	**b)** Make two logical expressions syntactically identical	Unification finds a substitution θ so that two expressions match exactly — the basis for resolution and backward chaining.
6	**c)** A contradiction — the	In resolution refutation,

Q	Correct Option	Justification
	original goal is proven	deriving □ from KB ∧ ¬Goal shows the negated goal is unsatisfiable, i.e., the goal is entailed.
7	**b)** Data-driven / bottom-up	Forward chaining starts from known facts and fires rules to derive new facts, moving "up" toward conclusions.
8	**b)** The goal and works backward to verify supporting facts	Backward chaining starts at the goal and recursively reduces it to sub-goals matched against known facts.
9	**d)** Distribute AND over OR	This is stated backwards — correct CNF requires distributing **OR over AND**. The other three (eliminate ↔, push ¬ inward, Skolemize) are genuine CNF-conversion steps.
10	**b)** Defining domain concepts, relations, and axioms formally in FOL	Knowledge engineering is the process of formally modeling a domain's objects, relations, and rules as FOL sentences.
11	**b)** Q must be true	Modus Ponens: from P -> Q and P, Q is validly derived.
12	**b)** Makes two terms identical with minimal variable binding	The MGU is the *least* constraining substitution that unifies two terms — more general than any other valid unifier.
13	**c)** Backward Chaining	Backward chaining is goal-directed, so it's efficient when a specific goal/query is already known, avoiding irrelevant derivations.
14	**b)** Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions	Propositional logic only has atomic true/false statements; FOL adds objects, predicates, and quantifiers (∀, ∃).
15	**b)** Adding ¬G to the KB and deriving a contradiction (empty clause □)	This is the definition of proof by refutation: assume the goal false, show the KB

Q	Correct Option	Justification
		becomes inconsistent.
16	**b)** Improving performance by examining examples or experience from the environment	Learning from observation means the agent improves its behavior based on the examples/experience it encounters.
17	**b)** Generalizing a hypothesis from specific training examples	Induction moves from specific observed instances to a general rule — the opposite direction of deduction.
18	**c)** A test on an attribute / feature	Internal nodes in a decision tree branch on a feature test; leaf nodes hold the class label/decision.
19	**c)** Uses prior domain knowledge to explain and generalize a single example	EBL uses background/domain theory to explain why one example is a positive instance, then generalizes that explanation — unlike inductive learning, which needs many examples.
20	**c)** All features are conditionally independent given the class label	This is the “naive” conditional-independence assumption that gives Naive Bayes its name.
21	**c)** Directly computing maximum likelihood estimates from observed data	With complete data (nothing hidden), parameters can be estimated directly by counting/MLE — no iterative estimation needed.
22	**c)** The dataset contains hidden (latent) variables that cannot be directly observed	EM alternates between estimating hidden variables (E-step) and re-estimating parameters (M-step) — needed precisely when some variables are unobserved.
23	**c)** Introduce non-linearity to enable learning of complex patterns	Without a non-linear activation function, a neural network collapses to a linear model regardless of depth.
24	**c)** The error is propagated backward from output to input layers to adjust weights	Backpropagation computes gradients of the loss w.r.t. weights by propagating error

Q	Correct Option	Justification
		backward through the network via the chain rule.
25	**c)** Finding the maximum-margin hyperplane that separates classes	SVMs choose the separating hyperplane that maximizes the margin between the closest points of each class (support vectors).
26	**c)** Operating in a high-dimensional feature space without explicitly computing coordinates	The kernel trick computes inner products in a higher-dimensional space implicitly via a kernel function, avoiding explicit feature mapping.
27	**c)** Cumulative reward received over time through interaction with the environment	RL agents choose actions to maximize expected long-run (cumulative/discounted) reward, not a one-shot accuracy metric.
28	**c)** A model-free, off-policy temporal difference learning method	Q-Learning learns action-values directly from experience (model-free), updates using TD error, and learns the optimal policy independent of the behavior policy (off-policy).
29	**c)** The model fits training data too closely and generalizes poorly to new data	Overfitting: excellent training performance but poor generalization to unseen/test data.
30	**c)** RL learns via trial-and-error with reward feedback; SL learns from labeled examples	This is the core distinction: RL has no labeled targets, only reward signals from interaction; SL trains directly on input–output pairs.

Answer Key (Quick Reference)

Q	Ans	Q	Ans	Q	Ans
1	b	11	b	21	c
2	d	12	b	22	c
3	b	13	c	23	c
4	b	14	b	24	c

Q	Ans	Q	Ans	Q	Ans
5	b	15	b	25	c
6	c	16	b	26	c
7	b	17	b	27	c
8	b	18	c	28	c
9	d	19	c	29	c
10	b	20	c	30	c

Total: 30/30 marks (1 mark per question)